📓 SOC Internship – Day 15 Notes

🔍 Log Correlation & Attack Chain Analysis

🎯 Objective

Understand how multiple log events are connected and how attackers perform **multi-step attacks**.

🧠 1. What is Log Correlation?

- Log correlation is the process of **combining multiple logs** to detect a **complete attack pattern**.
- Instead of analyzing logs individually, SOC analysts connect events to see the bigger picture.

⚠️ 2. Why Log Correlation is Important

- A **single log is not enough** to detect attacks
- Multiple related logs help reveal **hidden attack patterns**
- Enables SOC teams to identify **real security incidents**

🔗 3. Attack Chain Example

Step 1: Failed Logins

- Failed login from IP: 185.23.11.5 (multiple attempts)

Step 2: Successful Login

- Login success from same IP

Step 3: Suspicious Activity

- File accessed: /admin/config

📊 4. Analysis

- These are **not separate events** ❌
- Together, they form a **complete attack chain** ✅
- Indicates **possible account compromise**

🧩 5. SOC Correlation Thinking

SOC analysts ask:

- Are logs from the **same IP address**?
- Are events happening **close in time**?
- Is the behavior **unusual or suspicious**?

🚨 6. Detection Logic Example

IF multiple failed logins
AND successful login
AND suspicious activity
THEN alert: Possible account compromise

🏢 7. SOC Workflow

- **L1 (Level 1):** Detect alerts
- **L2 (Level 2):** Correlate events
- **L3 (Level 3):** Improve detection rules

🧪 8. Class Activity

Logs:

- Failed login (IP 1)
- Failed login (IP 1)
- Successful login (IP 1)
- File access

👉 **Conclusion:**
This is an **attack pattern**, not normal behavior.

9. Assignment

1. Explain log correlation
2. Identify attack chain from logs
3. Write full attack sequence